SME ERP SYSTEM

Small & Medium Enterprise Resource Planning

Comprehensive Project Documentation

Includes: System Overview • ERD • API Design • UI Structure

Version 1.0 | 2025

1. Project Overview

SME ERP System is a simplified Enterprise Resource Planning platform designed specifically for small and medium-sized businesses. It unifies core business operations — Sales, Inventory, Purchases, and Customer/Supplier management — into a single, easy-to-use platform, reducing the complexity typical of large ERP systems.

1.1 Problem Statement

Small businesses typically suffer from:

- Using scattered tools (Excel, paper, WhatsApp) with no centralized data
- Difficulty tracking sales, inventory levels, and purchase orders
- Frequent human errors due to manual data entry
- Poor decision-making from lack of unified reporting

1.2 Proposed Solution

The system provides a unified platform covering:

- Sales Management — quotations, orders, invoices
- Inventory Management — products, stock, warehouses
- Customer & Supplier Management
- Purchases Management — purchase orders, invoices
- Analytics Dashboard — real-time KPIs and alerts

1.3 Target Users

User Type	Description	Size
Small Businesses	Retail shops, distribution companies	10–50 employees
Service Companies	Agencies, consultancies, service providers	10–50 employees
Distributors	Wholesale and supply chain businesses	10–50 employees

2. System Architecture

2.1 Technology Stack

Layer	Technology	Status
Frontend	React + TypeScript + Tailwind CSS	✅Ready
Backend	Node.js / REST API (Modular Architecture)	□ In Development
Database	PostgreSQL	✅Ready
Auth	JWT + Role-Based Access Control (RBAC)	□ In Development
Document Mgmt	Reused from EnMS System	✅Ready

2.2 Backend Module Structure

REST API organized in modular architecture:

Module	Base Path	Responsibilities
Sales	/api/v1/sales	Quotations, Orders, Invoices
Inventory	/api/v1/inventory	Products, Stock, Warehouses
Purchases	/api/v1/purchases	Suppliers, PO, Purchase Invoices
Customers	/api/v1/customers	Customer profiles and history
Users	/api/v1/users	Authentication, roles, permissions
Reports	/api/v1/reports	Sales, inventory, purchase reports

2.3 User Roles

Role	Access Level	Modules
Admin	Full operational access	Dashboard, Sales, Inventory, Purchases, Customers, Reports, Documents, Config
SysAdmin	System management access	Users, Roles, Permissions, System Info, Jobs, Maintenance

3. Module Descriptions

3.1 Dashboard Module

Real-time operational overview showing:

- Total Sales (daily, weekly, monthly)
- Top-selling products
- Current inventory status with low-stock alerts
- Pending orders and recent transactions

3.2 Sales Module

Full sales lifecycle management:

Sub-Module	Description	Key Fields
Customers	Manage customer profiles	Name, contact, address, credit limit
Quotations	Create & send price quotes	Customer, items, qty, price, validity
Orders	Convert quotations to orders	Order #, status, delivery date
Invoices	Generate invoices from orders	Invoice #, amount, tax, payment status

Sales Workflow:

Customer → Quotation → Order → Invoice → Stock Update

3.3 Inventory Module

Stock and warehouse management:

Sub-Module	Description	Key Fields
Products	Product catalog management	SKU, name, category, unit, price
Stock	Current stock levels per location	Product, warehouse, quantity, min level
Warehouses	Manage multiple storage locations	Name, location, capacity

Stock Logic: Purchase → Stock Increase | Sale → Stock Decrease

3.4 Purchases Module

Sub-Module	Description	Key Fields

Suppliers	Supplier profiles	Name, contact, payment terms
Purchase Orders	Order goods from suppliers	Supplier, items, expected delivery
Purchase Invoices	Record received goods & payment	PO reference, amount, date received

3.5 Configuration Module

- User management (add/edit/deactivate users)
- Role & permission management
- Department configuration
- System settings & preferences

3.6 Reports Module

Report	Filters	Outputs
Sales Report	Date range, customer, product	Total sales, order count, top customers, chart
Inventory Report	Category, warehouse, date	Stock levels, movements, low-stock alerts
Purchase Report	Date range, supplier	Total purchases, PO status, supplier breakdown

4. Module Relationships & Workflows

4.1 Core Relationships

Module A	Relationship	Module B	Trigger
Sales	→ decreases stock	Inventory	When Order is confirmed
Purchases	→ increases stock	Inventory	When goods are received
Customers	→ linked to	Sales Orders	Each order has a customer
Suppliers	→ linked to	Purchase Orders	Each PO has a supplier
Users & Roles	→ controls access to	All Modules	Permission-based gating

4.2 Sales Workflow (Detailed)

1. Create or select Customer
2. Create Quotation with products and prices
3. Customer approves → Convert to Order
4. Generate Invoice from Order
5. Confirm Invoice → Auto-decrease Inventory

4.3 Purchase Workflow (Detailed)

6. Create or select Supplier
7. Create Purchase Order (PO) with items and quantities
8. Supplier delivers goods
9. Record receipt → Create Purchase Invoice
10. Confirm receipt → Auto-increase Inventory

5. Entity Relationship Diagram (ERD)

The following tables represent the complete database schema for the SME ERP System.

5.1 Core Tables

users

Column	Type	Constraints	Description
id	UUID	PK	Primary key
username	VARCHAR(100)	UNIQUE, NOT NULL	Login username
email	VARCHAR(255)	UNIQUE, NOT NULL	Email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt hashed password
role_id	UUID	FK → roles.id	Assigned role
department_id	UUID	FK → departments.id	Department reference
is_active	BOOLEAN	DEFAULT true	Account status
created_at	TIMESTAMP	DEFAULT NOW()	Creation date

roles

Column	Type	Constraints	Description
id	UUID	PK	Primary key
name	VARCHAR(100)	UNIQUE, NOT NULL	Role name (Admin, SysAdmin)
description	TEXT		Role description

permissions

Column	Type	Constraints	Description
id	UUID	PK	Primary key
role_id	UUID	FK → roles.id	Role reference
module	VARCHAR(100)	NOT NULL	Module name (sales, inventory…)
can_read	BOOLEAN	DEFAULT false	Read permission
can_write	BOOLEAN	DEFAULT false	Write permission
can_delete	BOOLEAN	DEFAULT false	Delete permission

5.2 Sales Tables

customers

Column	Type	Constraints	Description
id	UUID	PK	Primary key
name	VARCHAR(255)	NOT NULL	Customer name
email	VARCHAR(255)		Contact email
phone	VARCHAR(50)		Phone number
address	TEXT		Address
credit_limit	DECIMAL(12,2)	DEFAULT 0	Credit limit
is_active	BOOLEAN	DEFAULT true	Active status
created_at	TIMESTAMP	DEFAULT NOW()	Creation date

quotations

Column	Type	Constraints	Description
id	UUID	PK	Primary key
customer_id	UUID	FK → customers.id	Customer reference
quotation_number	VARCHAR(50)	UNIQUE, NOT NULL	Auto-generated number
status	ENUM	draft/sent/accepted/rejected	Current status
total_amount	DECIMAL(12,2)	NOT NULL	Total before tax
tax_amount	DECIMAL(12,2)	DEFAULT 0	Tax amount
valid_until	DATE		Expiry date
notes	TEXT		Additional notes
created_by	UUID	FK → users.id	Created by user
created_at	TIMESTAMP	DEFAULT NOW()	Creation date

quotation_items

Column	Type	Constraints	Description
id	UUID	PK	Primary key
quotation_id	UUID	FK → quotations.id	Quotation reference
product_id	UUID	FK → products.id	Product reference
quantity	DECIMAL(10,2)	NOT NULL	Ordered quantity

unit_price	DECIMAL(12,2)	NOT NULL	Price per unit
discount	DECIMAL(5,2)	DEFAULT 0	Discount %
line_total	DECIMAL(12,2)	COMPUTED	qty × price × (1-discount)

sales_orders

Column	Type	Constraints	Description
id	UUID	PK	Primary key
quotation_id	UUID	FK → quotations.id	Source quotation
customer_id	UUID	FK → customers.id	Customer reference
order_number	VARCHAR(50)	UNIQUE, NOT NULL	Auto-generated
status	ENUM	pending/confirmed/delivered/cancelled	Order status
delivery_date	DATE		Expected delivery
total_amount	DECIMAL(12,2)	NOT NULL	Total amount
created_at	TIMESTAMP	DEFAULT NOW()	Creation date

invoices

Column	Type	Constraints	Description
id	UUID	PK	Primary key
order_id	UUID	FK → sales_orders.id	Source order
customer_id	UUID	FK → customers.id	Customer reference
invoice_number	VARCHAR(50)	UNIQUE, NOT NULL	Auto-generated
status	ENUM	unpaid/partial/paid	Payment status
amount_due	DECIMAL(12,2)	NOT NULL	Total due
amount_paid	DECIMAL(12,2)	DEFAULT 0	Amount paid so far
due_date	DATE		Payment due date
issued_at	TIMESTAMP	DEFAULT NOW()	Issue date

5.3 Inventory Tables

products

Column	Type	Constraints	Description
id	UUID	PK	Primary key
sku	VARCHAR(100)	UNIQUE, NOT NULL	Stock keeping unit
name	VARCHAR(255)	NOT NULL	Product name
category_id	UUID	FK → categories.id	Product category
unit	VARCHAR(50)	NOT NULL	Unit of measure (pcs, kg, L)
purchase_price	DECIMAL(12,2)	DEFAULT 0	Cost price
selling_price	DECIMAL(12,2)	DEFAULT 0	Selling price
min_stock_level	DECIMAL(10,2)	DEFAULT 0	Reorder threshold
is_active	BOOLEAN	DEFAULT true	Active status

warehouses

Column	Type	Constraints	Description
id	UUID	PK	Primary key
name	VARCHAR(255)	NOT NULL	Warehouse name
location	TEXT		Physical address/location
is_active	BOOLEAN	DEFAULT true	Active status

stock

Column	Type	Constraints	Description
id	UUID	PK	Primary key
product_id	UUID	FK → products.id	Product reference
warehouse_id	UUID	FK → warehouses.id	Warehouse reference
quantity	DECIMAL(10,2)	NOT NULL, DEFAULT 0	Available quantity
reserved_qty	DECIMAL(10,2)	DEFAULT 0	Reserved for pending orders
updated_at	TIMESTAMP	DEFAULT NOW()	Last update

stock_movements

Column	Type	Constraints	Description
id	UUID	PK	Primary key
product_id	UUID	FK → products.id	Product reference
warehouse_id	UUID	FK → warehouses.id	Warehouse reference
type	ENUM	in/out/adjustment	Movement type
quantity	DECIMAL(10,2)	NOT NULL	Quantity moved
reference_type	VARCHAR(50)		sale_order/purchase/adjustment
reference_id	UUID		FK to source record
created_by	UUID	FK → users.id	User who triggered
created_at	TIMESTAMP	DEFAULT NOW()	Movement date

5.4 Purchases Tables

suppliers

Column	Type	Constraints	Description
id	UUID	PK	Primary key
name	VARCHAR(255)	NOT NULL	Supplier name
email	VARCHAR(255)		Contact email
phone	VARCHAR(50)		Phone number
address	TEXT		Address
payment_terms	INTEGER	DEFAULT 30	Days credit (0, 30, 60, 90)
is_active	BOOLEAN	DEFAULT true	Active status

purchase_orders

Column	Type	Constraints	Description
id	UUID	PK	Primary key
supplier_id	UUID	FK → suppliers.id	Supplier reference
po_number	VARCHAR(50)	UNIQUE, NOT NULL	Auto-generated PO number
status	ENUM	draft/sent/received/cancelled	PO status
expected_date	DATE		Expected delivery date
total_amount	DECIMAL(12,2)	NOT NULL	Total PO value
received_at	TIMESTAMP		Actual receipt timestamp
created_by	UUID	FK → users.id	Created by
created_at	TIMESTAMP	DEFAULT NOW()	Creation date

purchase_order_items

Column	Type	Constraints	Description
id	UUID	PK	Primary key
po_id	UUID	FK → purchase_orders.id	PO reference
product_id	UUID	FK → products.id	Product reference
quantity_ordered	DECIMAL(10,2)	NOT NULL	Ordered quantity

quantity_received	DECIMAL(10,2)	DEFAULT 0	Actually received
unit_price	DECIMAL(12,2)	NOT NULL	Price per unit
line_total	DECIMAL(12,2)	COMPUTED	qty × price

5.5 ERD Relationship Summary

Table	References	Type
users	roles, departments	Many-to-One
permissions	roles	Many-to-One
quotations	customers, users	Many-to-One
quotation_items	quotations, products	Many-to-One
sales_orders	quotations, customers	Many-to-One
invoices	sales_orders, customers	Many-to-One
stock	products, warehouses	Many-to-One (composite PK)
stock_movements	products, warehouses, users	Many-to-One
purchase_orders	suppliers, users	Many-to-One
purchase_order_items	purchase_orders, products	Many-to-One

6. API Design

All endpoints follow RESTful conventions. Base URL: /api/v1
Auth: Bearer JWT token required on all endpoints except login.

6.1 Authentication API

Method	Endpoint	Description	Auth Required
POST	/auth/login	Login with username + password → returns JWT	No
POST	/auth/logout	Invalidate current token	Yes
GET	/auth/me	Get current user profile	Yes
PUT	/auth/change-password	Change own password	Yes

6.2 Sales API — Customers

Method	Endpoint	Description	Body/Params
GET	/customers	List all customers	Query: search, page, limit
GET	/customers/:id	Get customer details	Param: id (UUID)
POST	/customers	Create new customer	name, email, phone, address, credit_limit
PUT	/customers/:id	Update customer	Any customer fields
DELETE	/customers/:id	Soft-delete customer	Param: id

6.3 Sales API — Quotations

Method	Endpoint	Description	Body/Params
GET	/sales/quotations	List quotations	Query: status, customer_id, date_from, date_to
GET	/sales/quotations/:id	Get quotation with items	Param: id
POST	/sales/quotations	Create quotation	customer_id, items[], valid_until, notes
PUT	/sales/quotations/:id	Update draft quotation	Any quotation fields
POST	/sales/quotations/:id/convert	Convert to Sales Order	Param: id
DELETE	/sales/quotations/:id	Delete draft quotation	Param: id

6.4 Sales API — Orders & Invoices

Method	Endpoint	Description	Notes
GET	/sales/orders	List orders	Filter by status, customer
GET	/sales/orders/:id	Get order details	
PUT	/sales/orders/:id/status	Update order status	Body: { status }
POST	/sales/orders/:id/invoice	Generate invoice from order	
GET	/sales/invoices	List invoices	Filter by status, customer
GET	/sales/invoices/:id	Get invoice details	
POST	/sales/invoices/:id/pay	Record payment	Body: { amount, payment_date }

6.5 Inventory API

Method	Endpoint	Description	Body/Params
GET	/inventory/products	List all products	Query: category, search, is_active
GET	/inventory/products/:id	Get product with stock levels	Param: id
POST	/inventory/products	Create product	sku, name, category_id, unit, prices, min_stock
PUT	/inventory/products/:id	Update product	Any product fields
GET	/inventory/stock	Get stock levels	Query: product_id, warehouse_id
POST	/inventory/adjustments	Manual stock adjustment	product_id, warehouse_id, qty, reason
GET	/inventory/movements	Stock movement history	Query: product_id, type, date range
GET	/inventory/low-stock	Products below min level	

6.6 Purchases API

Method	Endpoint	Description	Body/Params
GET	/suppliers	List suppliers	Query: search, is_active

POST	/suppliers	Create supplier	name, email, phone, address, payment_terms
PUT	/suppliers/:id	Update supplier	Any supplier fields
GET	/purchases/orders	List purchase orders	Query: status, supplier_id, date range
POST	/purchases/orders	Create purchase order	supplier_id, items[], expected_date
PUT	/purchases/orders/:id/receive	Mark order as received	items[]: { id, qty_received }
GET	/purchases/invoices	List purchase invoices	Query: po_id, status

6.7 Reports API

Method	Endpoint	Description	Query Params
GET	/reports/sales	Sales summary report	date_from, date_to, customer_id
GET	/reports/inventory	Inventory status report	category_id, warehouse_id
GET	/reports/purchases	Purchase summary report	date_from, date_to, supplier_id
GET	/reports/dashboard	Dashboard KPIs	period (today/week/month)

6.8 Standard API Response Format

Success Response:

```
{ "success": true, "data": { ... }, "meta": { "page": 1, "total": 100 } }
```

Error Response:

```
{ "success": false, "error": { "code": "NOT_FOUND", "message": "Resource not found"
} }
```

7. UI Structure per Module

7.1 Global Layout

Component	Description	Elements
Header	Top navigation bar	Logo, Language selector (AR/EN), Notifications, Profile menu
Sidebar	Left navigation panel	Module links, collapsible sub-menus, role-based visibility
Main Body	Content area	Breadcrumbs, Page title, Module content
Footer	Bottom bar	Version info, Quick links

7.2 Dashboard UI

Section	Component Type	Data Shown
KPI Row (top)	4 metric cards (Sales, Orders, Low Stock, Revenue)	Today / This Week / This Month
Sales Chart	Line/Bar chart	Sales trend over last 30 days
Top Products	Ranked list with progress bars	Top 5 by revenue or quantity
Recent Orders	Mini table	Last 10 orders with status badges
Alerts Panel	Warning cards (red/yellow)	Low stock items, overdue invoices

7.3 Customers UI

View	Components	Actions
List View	Searchable table with filters (status, date)	Add New, Edit, View, Delete
Detail View	Profile card + tabs (Orders, Invoices, History)	Edit, Deactivate
Create/Edit Form	Name, Email, Phone, Address, Credit Limit	Save, Cancel

7.4 Quotations UI

View	Components	Actions

List View	Table with status badges (Draft, Sent, Accepted, Rejected)	Create, View, Edit, Convert, Delete
Create Form	Customer picker, Item rows (product + qty + price + discount), Total summary	Save Draft, Send, Preview PDF
Detail View	Full quotation card with line items, totals, notes	Edit, Convert to Order, Print/Export

7.5 Sales Orders UI

View	Components	Actions
List View	Table with status filter (Pending, Confirmed, Delivered, Cancelled)	View, Change Status, Create Invoice
Detail View	Order header (customer, date, delivery date) + Items table	Update Status, Generate Invoice, Print

7.6 Invoices UI

View	Components	Actions
List View	Table with payment status (Unpaid, Partial, Paid), overdue highlighted in red	View, Record Payment, Send Email
Detail View	Invoice card (order ref, customer, amounts, due date) + Payment history	Record Payment, Print PDF

7.7 Inventory — Products UI

View	Components	Actions
List View	Card grid or table with stock indicator (green/yellow/red)	Add Product, Edit, View Stock
Detail View	Product info card + Stock by Warehouse table + Movement history log	Edit, Adjust Stock
Create/Edit Form	SKU, Name, Category, Unit, Purchase/Selling Price, Min Stock Level	Save, Cancel

7.8 Purchase Orders UI

View	Components	Actions
List View	Table with status (Draft, Sent, Received, Cancelled), supplier name, date	Create PO, View, Receive Goods
Create Form	Supplier picker, expected date, Items table (product + qty + price)	Save Draft, Send to Supplier
Receive Goods View	PO items list with quantity_received input per item	Confirm Receipt → Updates Stock

7.9 Reports UI

Report	Filter Panel	Output Components
Sales Report	Date range picker, Customer dropdown	Summary cards + Line chart + Orders table
Inventory Report	Category filter, Warehouse dropdown	Stock level table with color indicators
Purchase Report	Date range, Supplier dropdown	PO summary + Supplier breakdown bar chart

7.10 Configuration UI

Section	Components	Actions
Users	User list table with role badges	Add User, Edit, Deactivate, Reset Password
Roles & Permissions	Role cards + Permission matrix grid (modules × actions)	Edit Permissions per Role
Departments	Simple list/table	Add, Edit, Delete

8. Reuse from Existing EnMS System

EnMS Module	ERP Equivalent	Status	Changes Required
users	users	✅Reuse as-is	None
permissions	permissions	✅Reuse as-is	Add ERP modules to permission list
documentManagement	documents	✅Reuse as-is	Minor UI relabeling
dashboard	dashboard	□ Modify	Replace EnMS KPIs with ERP KPIs
measurements	inventory	□ Repurpose	Full redesign of data model
pipelines	order flow	□ Repurpose	Adapt workflow engine
commitment	(removed)	❌Remove	Not applicable to ERP
energy analysis	(removed)	❌Remove	Not applicable to ERP
ISO modules	(removed)	❌Remove	Not applicable to ERP

9. Limitations & Future Improvements

9.1 Current Limitations

Feature	Status	Reason
Advanced Accounting (GL, AP, AR)	✕Not included	Out of scope for v1.0
HR / Payroll Management	✕Not included	Out of scope for v1.0
AI / Advanced Analytics	✕Not included	Complexity outside MVP
Mobile App	✕Not included	Web-first approach

9.2 Future Roadmap

11. Accounting Module — General Ledger, AP, AR, Financial Statements
12. Mobile Application — iOS & Android companion app
13. Push Notifications — Low stock alerts, invoice reminders
14. Advanced Analytics — Sales forecasting, trend analysis
15. Multi-currency Support
16. E-commerce Integration — Sync orders from online stores

10. Project Summary

Attribute	Value
System Name	SME ERP System
Version	1.0
Target Users	Small businesses (10–50 employees)
Frontend	React + TypeScript + Tailwind CSS
Backend	Node.js REST API (Modular)
Database	PostgreSQL
Auth	JWT + RBAC
Core Modules	Dashboard, Sales, Inventory, Purchases, Customers, Reports, Config
Total DB Tables	18+ tables with full relational structure
API Endpoints	35+ RESTful endpoints
Roles	Admin, SysAdmin

— End of Document —